

12	B
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20	C
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Question	Marking guidance	Mark	Comments
10.1	The (magnitude of the) momentum of the block is smaller (than that of the object) ✓ The two momenta are in different directions wtte ✓	2	Not just that the magnitudes are different Accept that “they” are in different directions
10.2	Attempts to use (for the object) $mg - T = ma$ ✓ $T = 3.0(3) \text{ (N)}$ ✓	2	Expect to see $0.4g - T = 0.4a$ but accept incorrect equation with 3 terms – possible with incorrect sign
10.3	Use of $v^2 = u^2 + 2as$ ✓ $v = 1.42$ to at least 3 sf (m s^{-1}) ✓	2	Credit use of $s = \frac{1}{2}gt^2$ with $v = at$
10.4	$mg \sin 30$ or $mg \cos 60$ 1.23 (N) ✓	2	

Question	Marking guidance	Mark	Comments
10.5	Adds frictional force to component of weight parallel to slope: $1.25 + 1.23 (2.48)$ or $1.25 + \text{candidate's}$ 10.4 ✓ Use of $E_k = \frac{1}{2}mv^2$ and work = force x distance or $F = ma$ and $v^2 = u^2 + 2as$ (or other legitimate combination of equations of motion) ✓ $d = 0.10(2) \text{ m}$ ✓	3	For equⁿ of motion method expect to see $a = 9.905 \text{ m s}^{-2}$ Use of $v = 1.4$ gives $d = 0.099 \text{ m}$ for full marks

Question	Marking guidance	Mark	Comments
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17	A
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18	D
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19	D
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21	A
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17	D	5 N s
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Question	Answers	Additional comments/Guidelines	Mark	AO
07.1	Uses $E_p = mgh$ ✓ 180 (m) ✓	In MP1 allow one transcription error for value of mass or E_p. If more than 2 sf only accept answers that round to 176 (m)	2	1 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
07.2	Uses $E_k = \frac{1}{2}mv^2$ to attempt to find initial kinetic energy ✓ Uses final KE = their initial KE + work done + GPE ✓ 81(.4) (m s⁻¹) ✓	Look for 344 J In MP2 tolerate omission of one of the terms on the right hand side of the equation	3	2 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
07.3	Idea that not all of the work done by the falcon goes to increasing its kinetic energy ✓ (wings disturb the air so) some of the work will increase kinetic energy of the air OR Work is done heating the air OR Energy is transferred due to air resistance/drag ✓	Allow idea that work is done against air resistance/drag	2	1 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
07.4	Idea that before and after an interaction momentum is the same for a closed system/ in the absence of external forces ✓	Condone collision for interaction	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
07.5	Uses momentum = mv ✓ Recognisable conservation of momentum equation with some substitution Or one initial momentum calculated correctly (27.5 or 2.16) ✓ $23(.2) \text{ (m s}^{-1}\text{)} \checkmark$		3	2 × AO2 1 × AO3
Total			11	

Question	Answers	Additional comments/Guidelines	Mark	AO
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14	C	moment $\text{kg m}^2 \text{s}^{-2}$	AO1
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19	A	140 N s	AO3
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Question	Answers	Additional comments/Guidelines	Mark	AO
08.1	read off from section A of graph and divide by g ✓	expect to see 56.1 kg but condone 2 sf	1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
08.2	counts squares in the hump (above the weight line) = 11.5–13 big squares ✓₁ determines area of one square = 12.5 kg m s⁻¹ ✓₂ Alternative: calculation of a rectangle: height tolerance of (900 to 1100) ✓₁ width tolerance of (0.12 to 0.16) ✓₂ momentum = 140–163 (kg m s⁻¹) (must be at least 2 sf) ✓₃	condone MP1 (either alternative) for including below the weight line MP3 is dependent on MP1 and MP2 working required for final mark	3	AO3 × 3

Question	Answers	Additional comments/Guidelines	Mark	AO
08.3	use of $p = mu$ (to give $u = 2.5\text{--}3.0 \text{ m s}^{-1}$) ✓	ecf from 08.2	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
08.4	uses $v = u + at$ (or alternative suitable suvat equations) to give time to reach peak height ✓ doubles time to determine total time of flight (expect 0.50 to 0.60 s) ✓	ecf from 08.3 give both MP1 and MP2 for use of $s = ut + \frac{1}{2} at^2$ to give the total time directly	2	AO2 × 2

Question	Answers	Additional comments/Guidelines	Mark	AO
08.5	finds time of flight from graph = 0.54 to 0.56 s ✓	some working or annotation of the graph required 2 sf only	1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
08.6	idea that determination in 08.5 has a smaller percentage uncertainty (or reverse argument) because estimating the area under the curve (e.g. counting squares) is likely to be inaccurate ✓		1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
08.7	statement of Newton's 2 nd law in terms of change of momentum ✓ change of momentum/impulse is the same ✓ increasing (contact) time decreases the (average) force ✓	law must be named and stated	3	AO1 × 3

Total			12	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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14	B	speed distance	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
05	(Estimation of mass = 0.05 kg to 0.5 kg) Use of $v^2 = u^2 + 2as$ OR $v = \sqrt{2gh}$ ✓ Use of $p = mv$ with their v and their m ✓ Answer that rounds to value between 0.4 and 4 (kg m s^{-1}) 1 or 2 sf ✓	Award MP1 for an unsupported value of v of 8 m s^{-1} or 10 m s^{-1} Award both MP1 and MP2 for $mgh = \frac{p^2}{2m}$ from energy approach. Do not award MP3 if incorrect method for MP1 or MP2 seen.	3	AO2 × 3
Total			3	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
02.1	It has magnitude and direction ✓	allow the product of a scalar / mass and a vector (velocity) is a vector	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
02.2	Calculates total initial momentum $-14.4 \text{ kg m s}^{-1}$ ✓ Idea that momentum is conserved in the collision ✓ Truck X must be moving to the left so that total momentum after the collision is (also) to the left OR Truck X cannot be moving to the right otherwise net momentum would be to the right / momentum not conserved ✓	Max 2 if there is the assumption that X and Y move together after the collision Allow a clear indication of direction Allow if v is calculated there is a change in sign and a conclusion that the sign change shows change in direction	3	AO2
Total			4	

Question	Answers	Additional comments/Guidelines	Mark	AO
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17	B	$\Delta p \propto F$	$\Delta E_k \propto F^2$		AO1
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16	D	0 v	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
05	Attempt to use conservation of momentum ✓ $v = 3.6(3)$ seen ✓ Their final kinetic energy correctly calculated ✓ Initial and calculated final kinetic energy compared and a consistent conclusion given ✓	eg $(8.4 \times 4.1) + (6.3 \times 2.7) = (8.4 \times 3.4) + 6.3v$ final = 90.1 J	4	1 × AO1 2 × AO2 1 × AO3
Total			4	

Question	Answers	Additional comments/Guidelines	Mark	AO
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13	C	displacement, force, momentum	AO1
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14	D	$1.2 \times 10^{-2} \text{ kg m s}^{-1}$	AO2
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23	B	momentum.	AO1
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Question	Answers	Additional comments/Guidelines	Mark	AO
04	Use of $\frac{1}{2}mv^2$ to find E_k of proton(s) ✓ Converts their proton energy to MeV or electron / positron energy to J ✓ Correct total energy AND Compares E_k of proton(s) to rest energy and makes statement, no ✓	$\frac{1}{2} \times 1.67 \times 10^{-27} \times (1 \times 10^6)^2 = 8.36 \times 10^{-16}$ $\frac{1}{2} \times 1.67 \times 10^{-27} \times (2 \times 10^6)^2 = 3.35 \times 10^{-15}$ $8.18 \times 10^{-14} \text{ J OR } 1.64 \times 10^{-13} \text{ J}$ $4.18 \times 10^{-15} \text{ J OR } 0.026 \text{ MeV}$ Accept comparison to single electron	3	AO2
Total			3	

Question	Answers	Additional comments/Guidelines	Mark	AO
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17	A			AO1
		conserved	not conserved	

18	A	$240 \times 10^{-3} \text{ N s}$	AO2
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